Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

	Section A		**2*5=10**
1)	Explain the process of requirement elicitation and its importance in developing an ethical software product.	**2**	(CO1, L2)
2)	Describe the significance of database normalization and its different normal forms.	**2**	(CO2, L2)
3)	Explain the significance of different database management system layers and describe their functions.	**2**	(CO2, L2)
4)	Explain the process of creating a view in a relational database and illustrate it using DDL commands.	**2**	(CO3, L2)
5)	Explain and briefly describe the significance of different layers in software architecture.	**2**	(CO1, L2)
	Section B		**5*4=20**
6)	Analyze the differences between DDL and DML commands, providing examples of each to illustrate their usage in constructing a database project.	**5**	(CO3, L4)
7)	Analyze the process of creating views in a relational database management system, comparing their usage in a library management system versus a sales inventory system. Provide examples to illustrate your answer.	**5**	(CO1, L4)
8) a)	Analyze the importance of effective requirement elicitation in ensuring database quality and discuss potential issues that can arise if this process is not conducted properly. Provide an example to illustrate your points.	**5**	(CO2, L4)
	OR		
b)	Analyze the role of transaction management within a database system, explaining how it ensures data integrity and discussing its significance through an example.	**5**	(CO2, L4)
9)	Discuss the key steps involved in software quality assurance during the development lifecycle and explain how these steps contribute to ensuring the ethical development of a software product.	**5**	(CO1, L2)

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Section C

10*2=20

10)	Construct a relational database management system (RDBMS) architecture and explain how each layer interacts with the others. Additionally, discuss the importance of proper interaction between these layers in maintaining data integrity and ensuring efficient query processing.	**10**	(CO2, L3)
11) a)	Construct a database schema for a Library Management System using a relational model. Demonstrate your design by explaining how you have ensured referential integrity between tables. Additionally, show how to create and manipulate views that provide specific user perspectives (e.g., view of books available in a particular section or borrowed by a specific reader).	**10**	(CO1, L3)
	OR		
b)	Construct an ethical requirement elicitation process for a new e-learning platform and explain its importance. Additionally, determine potential ethical issues that could arise during this process and propose strategies to mitigate these issues.	**10**	(CO1, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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